

MMA LAB QUIZ
UPDATED MASTER MCQ NOTE

CSE 306 - Microprocessor and Microcontroller Sessional

Strictly filtered to the uploaded lecture/lab syllabus

366 UNIQUE MCQs | All previous 226 preserved + 140 new | New-note + web-inspired + code traps

Expanded edition: all previous MCQs + new MMA quiz notes + tricky code MCQs

How this master note was built

1. The uploaded slides and lab sheets define the syllabus boundary. Questions that require concepts not present in those files were excluded.
2. GeeksforGeeks, TutorialsPoint and Sanfoundry were used to cross-check standard 8086 terminology and to identify common MCQ patterns. Their questions were not copied verbatim; relevant ideas were rewritten and filtered against the class material.
3. Duplicate and near-duplicate prompts were removed. Many items are deliberately "trap-style": size mismatch, implicit registers, flag behavior, register-vs-memory, active-low display logic, and address/port mix-ups.
4. When a web page contains extra material outside the class slides (for example advanced 8255 modes), that material was not imported into the question bank.
5. This expanded edition preserves all 226 previous MCQs and adds new questions from the 35-page "MMA quiz" note, with extra emphasis on trainer-kit workflow, LCD program tracing, dot-matrix code, machine-code reading, and exam-style code traps.

Syllabus coverage map

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High-risk traps to memorize before the quiz

- 8086: 16-bit CPU, 20-bit address bus, 1 MB address space.
- Physical address = segment x 10H + offset.
- CL is the register used for variable shift/rotate counts in the taught syntax.
- TEST performs AND for flags but does not store the result; ZF=1 when that result is zero.
- LOOP automatically decrements CX and repeats while CX != 0.
- PUSH decrements SP by 2; POP increments SP by 2.
- Byte MUL: AX = AL x source. Word MUL: DX:AX = AX x source.
- Byte DIV: AX / src -> quotient AL, remainder AH. Word DIV: DX:AX / src -> quotient AX, remainder DX.
- Unsigned divide extension: clear AH/DX. Signed divide extension: CBW/CWD.
- Rotate preserves the bit pattern after a full-width rotation; logical shift discards bits.
- DOS INT 21H: AH=1 input -> AL, AH=2 character output from DL, AH=9 string output via DX and \$ terminator.
- 7-segment lab: common anode -> 0 lights a segment. Discrete LED lab: common cathode -> 1 lights an LED.
- Dot matrix: Port A green, Port B red, Port C columns; A+B gives orange.

- LCD BF=1 means busy; do not write the next instruction until BF=0.
- LCD line starts: 80H first line, C0H second line.

A. 8086 Architecture, Registers, Memory & Program Structure

1. The 8086 is described in the slides as a:

- A. 8-bit microprocessor
- B. 16-bit microprocessor
- C. 20-bit microprocessor
- D. 32-bit microprocessor

Answer: B | It can work on 16 bits at a time.

2. How wide is the 8086 address bus in the course slides?

- A. 8 bits
- B. 16 bits
- C. 20 bits
- D. 32 bits

Answer: C | A 20-bit address bus can address 2^{20} byte locations.

3. The maximum directly addressable memory implied by a 20-bit address bus is: **[TRAP]**

- A. 64 KB
- B. 256 KB
- C. 1 MB
- D. 16 MB

Answer: C | 2^{20} bytes = 1,048,576 bytes = 1 MB.

4. Which pair are the two major functional units of the 8086 CPU?

- A. ALU and RAM
- B. EU and BIU
- C. CS and DS
- D. AX and BX

Answer: B | The CPU is divided into the Execution Unit (EU) and Bus Interface Unit (BIU).

5. Which unit contains the ALU and performs arithmetic/logical operations?

- A. BIU
- B. EU
- C. ROM
- D. Stack segment

Answer: B | The EU contains the ALU and works with general registers.

6. Which unit communicates between the EU and memory/I/O?

- A. BIU
- B. ALU
- C. Flag unit
- D. Stack pointer

Answer: A | The BIU handles bus communication with RAM and I/O.

7. Which sequence best matches instruction execution in the slides?

- A. Decode -> Fetch -> Execute
- B. Fetch -> Execute -> Decode
- C. Fetch -> Decode -> Execute

D. Execute -> Fetch -> Decode

Answer: C | The slides list Fetch, Decode, Execute in that order.

8. How many total 16-bit registers are counted in the introductory slide?

- A. 8
- B. 12
- C. 14
- D. 16

Answer: C | The slide groups 4 data + 9 address + 1 flag register = 14.

9. Which register is most directly associated with loop counting?

- A. AX
- B. BX
- C. CX
- D. DX

Answer: C | CX is the count register and is used by LOOP.

10. Which register is emphasized for multiplication, division and I/O?

- A. AX
- B. BX
- C. SI
- D. BP

Answer: A | AX is the accumulator and is used for multiplication, division and I/O.

11. Which data register is also described as a base register?

- A. AX
- B. BX
- C. CX
- D. DX

Answer: B | BX is the Base Register.

12. Which data register is used with multiplication/division and output according to the register slide?

- A. AX only
- B. BX
- C. CX
- D. DX

Answer: D | DX is used in multiplication, division and output, especially for wider results/dividends.

13. Which list contains only segment registers?

- A. CS, DS, SS, ES
- B. SP, BP, IP, SI
- C. AX, BX, CX, DX
- D. SI, DI, CS, IP

Answer: A | The four segment registers are CS, DS, SS and ES.

14. Which list contains the pointer registers shown in the slides?

- A. SP, BP, IP

- B. SI, DI, IP
- C. CS, DS, ES
- D. AX, BX, CX

Answer: A | SP, BP and IP are listed as pointer registers.

15. Which are the index registers?

- A. AX and BX
- B. SI and DI
- C. SP and BP
- D. CS and DS

Answer: B | SI and DI are Source Index and Destination Index registers.

16. The 8086 physical address is formed by: **[TRAP]**

- A. segment + offset
- B. segment x 10H + offset
- C. offset x 10H + segment
- D. segment x 100H + offset

Answer: B | Shift the segment left 4 bits (multiply by 10H), then add offset.

17. What is the physical address of 1000H:0010H?

- A. 10010H
- B. 10001H
- C. 10100H
- D. 00110H

Answer: A | $1000H \times 10H = 10000H$; $+0010H = 10010H$.

18. Two different segment:offset pairs can refer to the same physical address because:

- A. segments are 8-bit
- B. segments can overlap
- C. offset is always zero
- D. BIU ignores segment values

Answer: B | Segmented addressing allows overlapping logical segments.

19. Which segment normally contains executable instructions?

- A. DS
- B. SS
- C. CS
- D. ES

Answer: C | CS is the Code Segment.

20. Which segment is used for stack data?

- A. CS
- B. DS
- C. SS
- D. ES

Answer: C | SS is the Stack Segment.

21. In the SMALL memory model used in the slides, the program has:

- A. many code and many data segments
- B. one code and one data segment
- C. one code and many data segments
- D. many code and one data segment

Answer: B | SMALL uses one code segment and one data segment, each up to 64 KB.

22. .STACK 100H reserves how many bytes? [TRAP]

- A. 100 bytes
- B. 128 bytes
- C. 256 bytes
- D. 512 bytes

Answer: C | 100H hexadecimal equals 256 decimal bytes.

23. What is the purpose of MAIN ENDP?

- A. Terminate DOS
- B. Mark the end of the MAIN procedure
- C. Initialize DS
- D. Declare stack size

Answer: B | PROC/ENDP delimit a procedure definition.

24. Why is END MAIN still needed after MAIN ENDP?

- A. It defines the data segment
- B. It tells the assembler the source end and entry point symbol
- C. It clears AX
- D. It creates a far procedure

Answer: B | END MAIN is an assembler directive marking the end and entry symbol; ENDP only closes the procedure.

25. Which memory model allows many code segments but only one data segment?

- A. SMALL
- B. MEDIUM
- C. COMPACT
- D. TINY

Answer: B | MEDIUM allows code >64 KB using many code segments while data stays <=64 KB.

B. Number Syntax, Variables, Basic Instructions & DOS I/O

26. Which is a valid binary literal according to the slides?

- A. 10110111
- B. 10110111b
- C. 0xB7
- D. B710

Answer: B | Binary literals end with b.

27. Which hexadecimal literal is valid?

- A. DAh
- B. 0DAh
- C. DA
- D. hDA

Answer: B | A hex literal beginning with A-F must be prefixed by a digit such as 0.

28. The literal 10110111 with no suffix is treated as:

- A. Binary
- B. Hexadecimal
- C. Decimal
- D. ASCII

Answer: C | Without b or h, the slide treats it as decimal.

29. Which variable name is illegal?

- A. @1
- B. AbC\$?
- C. ?
- D. 1Var

Answer: D | A variable name cannot start with a digit.

30. Assembly variable names in the slides are:

- A. Case-sensitive
- B. Case-insensitive
- C. Digits only
- D. Required to begin with @

Answer: B | The naming slide states case-insensitive.

31. DB defines a:

- A. 4-bit nibble
- B. 8-bit byte
- C. 16-bit word
- D. 32-bit double word

Answer: B | DB means Define Byte = 8 bits.

32. DW defines:

- A. 8 bits
- B. 16 bits
- C. 32 bits

D. 64 bits

Answer: B | DW means Define Word = 16 bits.

33. DD and DQ correspond to:

- A. 16 and 32 bits
- B. 32 and 64 bits
- C. 64 and 128 bits
- D. 8 and 16 bits

Answer: B | DD is 32 bits and DQ is 64 bits.

34. Why is VAR DB 320 invalid in the slide?

- A. 320 is negative
- B. 320 does not fit in an 8-bit byte
- C. DB requires binary only
- D. 320 is a register name

Answer: B | A byte can hold only an 8-bit value; 320 is too large.

35. Which instruction format is correct?

- A. Operand Opcode Comment
- B. Opcode Operands ;Comment
- C. Comment Operand Opcode
- D. Opcode ;Operand only

Answer: B | The slide shows opcode, operands, then semicolon comment.

36. What does MOV DEST, SRC do?

- A. DEST = DEST + SRC
- B. SRC = DEST
- C. DEST = SRC
- D. Swaps DEST and SRC

Answer: C | MOV copies the source value into destination.

37. Which MOV is illegal under the course restriction?

- A. MOV AH, DL
- B. MOV var1, AL when var1 is DB
- C. MOV var1, var2
- D. MOV AX, var1 when var1 is DW

Answer: C | The slides disallow both source and destination being memory variables at the same time.

38. Which MOV is illegal because of size mismatch?

- A. MOV AH, DL
- B. MOV AX, BX
- C. MOV CL, 1000000000000b
- D. MOV AL, 3

Answer: C | CL is 8-bit; that immediate does not fit the shown 8-bit use.

39. Which XCHG form is illegal?

- A. XCHG AH, DL
- B. XCHG var1, AL

- C. XCHG AX,30
- D. XCHG AX,BX

Answer: C | XCHG exchanges registers/memory, not an immediate constant.

40. ADD AL,BH means:

- A. $AL = BH$
- B. $AL = AL + BH$
- C. $BH = AL + BH$
- D. $AL = AL - BH$

Answer: B | ADD stores the sum in the destination.

41. SUB AL,BH means:

- A. $AL = AL - BH$
- B. $BH = BH - AL$
- C. $AL = BH - AL$
- D. Only flags change

Answer: A | SUB subtracts source from destination and stores the result in destination.

42. NEG AL performs:

- A. $AL = AL + 1$
- B. $AL = -AL$
- C. $AL = 0$
- D. $AL = \text{NOT } AL$ only

Answer: B | NEG forms the arithmetic negative (two's complement) of its operand.

43. INC AL is equivalent to:

- A. $AL = AL + 1$
- B. $AL = AL - 1$
- C. $AL = 1$
- D. $AL = AL + CF$

Answer: A | INC increments by one.

44. For DOS single-character input with INT 21H, AH should contain:

- A. 01H
- B. 02H
- C. 09H
- D. 4CH

Answer: A | AH=01H selects single-character input in the course slide.

45. After INT 21H with AH=1, the input character is returned in:

- A. AH
- B. AL
- C. DL
- D. DX

Answer: B | The input character is stored in AL.

46. For single-character output with INT 21H, the character must be placed in:

- A. AL

- B. AH
- C. DL
- D. DH

Answer: C | AH=2 selects output, and DL holds the character.

47. For DOS string output in the slide, AH should be:

- A. 1
- B. 2
- C. 9
- D. 10

Answer: C | AH=9 is the string-output service.

48. A string printed by the shown AH=9 service must end with:

- A. NUL
- B. CR
- C. \$
- D. #

Answer: C | The slide stores strings as DB data terminated by \$.

49. Before AH=9 string output, the string address is loaded into:

- A. AX using MOV
- B. DX using LEA
- C. CX using XCHG
- D. SP using PUSH

Answer: B | LEA DX,STRING loads the effective address into DX.

50. Why does the slide warn you to save a desired AL value before INT 21H? [TRAP]

- A. INT 21H always clears DS
- B. AL may be changed by the DOS service
- C. DX is always destroyed
- D. SP becomes zero

Answer: B | The slide explicitly warns that AL can change after INT 21H.

C. Control Flow, CMP, Conditional Jumps & Loops

51. Which instruction performs an unconditional transfer to a label?

- A. CMP
- B. JMP
- C. JE
- D. LOOP

Answer: B | JMP transfers control without testing a condition.

52. CMP is mainly used to:

- A. Store a subtraction result
- B. Update flags for a comparison
- C. Swap operands
- D. Push values

Answer: B | CMP compares values by subtraction-like flag updates without keeping the arithmetic result.

53. JE means:

- A. Jump if even
- B. Jump if equal
- C. Jump if error
- D. Jump if empty

Answer: B | JE branches when the comparison indicates equality.

54. JNE means:

- A. Jump if not equal
- B. Jump if negative
- C. Jump if no error
- D. Jump if no carry

Answer: A | JNE branches when values are not equal.

55. Which jump means greater?

- A. JG
- B. JL
- C. JGE
- D. JLE

Answer: A | JG is Jump if Greater.

56. Which jump means greater or equal?

- A. JG
- B. JGE
- C. JL
- D. JLE

Answer: B | JGE is Jump if Greater or Equal.

57. Which jump means less?

- A. JG
- B. JGE
- C. JL

D. JLE

Answer: C | JL is Jump if Less.

58. Which jump means less or equal?

- A. JG
- B. JGE
- C. JL
- D. JLE

Answer: D | JLE is Jump if Less or Equal.

59. JXCZ is used when:

- A. AX is zero
- B. BX is zero
- C. CX is zero
- D. CF is zero

Answer: C | JXCZ checks whether CX is zero.

60. JC and JNC test which flag?

- A. Zero
- B. Carry
- C. Sign
- D. Parity

Answer: B | JC jumps if carry; JNC jumps if no carry.

61. JO and JNO test:

- A. Overflow
- B. Zero
- C. Parity
- D. Sign

Answer: A | JO/JNO correspond to Overflow flag set/clear.

62. JP and JNP test:

- A. Pointer
- B. Parity
- C. Procedure
- D. Page

Answer: B | JP/JNP branch on the Parity flag.

63. JS and JNS test:

- A. Stack
- B. Sign
- C. Segment
- D. Shift

Answer: B | JS/JNS branch on the Sign flag.

64. The LOOP instruction uses which register automatically?

- A. AX
- B. BX

C. CX

D. DX

Answer: C | LOOP uses CX as its count register.

65. What does LOOP do to CX each iteration?

A. Increments it

B. Decrements it

C. Sets it to zero immediately

D. Rotates it

Answer: B | LOOP decrements CX and repeats while CX is not zero.

66. If CX starts at 5 and the loop body contains only LOOP TOP at the end, the body executes:

A. 4 times

B. 5 times

C. 6 times

D. Until CF becomes 0

Answer: B | The body executes once for each initial count value down to 1.

67. Why is the simple do-while translation guaranteed to execute its body at least once?

A. CMP is before the body

B. The condition is checked after the body

C. CX begins at zero

D. JMP always exits

Answer: B | A do-while checks its condition after executing the body.

68. To imitate a normal for-loop that might execute zero times, the condition should be checked:

A. Only after INC

B. Before entering/executing the body

C. Only after LOOP

D. Never

Answer: B | The slide's for-loop translation compares before the body so it can skip it.

69. In an assembly implementation of logical AND (&&), the first failed condition should usually:

[TRAP]

A. Still evaluate all code as true

B. Jump to the false/next branch

C. Set CF and continue true path

D. Call a procedure

Answer: B | Both conditions must be true, so any failure exits the true path.

70. In an assembly implementation of logical OR (|), a true first condition can usually:

A. Immediately enter the true branch

B. Force evaluation to false

C. Clear all flags

D. Skip the true branch

Answer: A | Only one OR term needs to be true.

D. Logical Operations, Bit Masks, Shift/Rotate & Binary/Hex I/O

71. AND DEST, SRC performs:

- A. Bitwise AND and stores in DEST
- B. Bitwise OR
- C. Only flag checking with no stored result
- D. Arithmetic addition

Answer: A | AND stores the bitwise AND result in destination.

72. OR is most useful for:

- A. Clearing selected bits
- B. Setting selected bits
- C. Dividing by 2
- D. Comparing signed numbers

Answer: B | OR with a 1 mask forces selected bits to 1.

73. AND with a 0 in a mask position will:

- A. Set that destination bit
- B. Clear that destination bit
- C. Toggle that bit
- D. Rotate that bit

Answer: B | Any bit AND 0 becomes 0.

74. XOR with 1 on a selected bit will:

- A. Keep it unchanged
- B. Clear it
- C. Toggle it
- D. Set CF only

Answer: C | XOR 1 complements that bit.

75. XOR with 0 on a bit will:

- A. Toggle it
- B. Leave it unchanged
- C. Always set it
- D. Always clear it

Answer: B | $X \text{ XOR } 0 = X$.

76. A common way to clear a register using only XOR is:

- A. XOR AL, 1
- B. XOR AL, AL
- C. XOR AL, 0FFH
- D. XOR AL, 32

Answer: B | Any value XOR itself becomes zero.

77. TEST differs from AND mainly because TEST:

- A. Does not perform a bitwise AND
- B. Does not store the AND result
- C. Never changes flags

D. Requires only one operand

Answer: B | TEST computes an AND for flag effects but does not store the result.

78. If TEST A,B produces an all-zero result, ZF becomes:

- A. 0
- B. 1
- C. Undefined
- D. Equal to CF

Answer: B | The Zero Flag is set when the result is zero.

79. To test whether AL is odd, the course uses:

- A. TEST AL,1
- B. TEST AL,2
- C. OR AL,1
- D. SHR AL,8

Answer: A | The least-significant bit determines odd/even.

80. After TEST AL,1, JZ EVEN is taken when:

- A. LSB=1
- B. LSB=0
- C. CF=1
- D. AL is negative

Answer: B | If the LSB is 0, the AND result is zero and ZF=1.

81. NOT performs:

- A. Two's complement
- B. One's complement
- C. Addition by 1
- D. Logical shift right

Answer: B | NOT flips all bits and does not add one.

82. NEG performs:

- A. One's complement only
- B. Two's complement arithmetic negation
- C. Rotate through carry
- D. Bitwise AND

Answer: B | NEG is two's complement: invert then add 1.

83. According to Week 9, which instruction does NOT affect flags?

- A. NEG
- B. NOT
- C. TEST
- D. AND

Answer: B | The slide says NOT does not affect any flags; NEG affects several.

84. SHL BL,1 sends the original:

- A. LSB to CF
- B. MSB to CF

- C. CF to MSB
- D. Zero Flag to LSB

Answer: B | One left shift moves the original MSB into CF.

85. SHR BL,1 sends the original:

- A. MSB to CF
- B. LSB to CF
- C. CF to LSB
- D. Parity flag to MSB

Answer: B | One right shift moves the original LSB into CF.

86. The vacant bit inserted by logical SHL is:

- A. 1 at MSB
- B. 0 at LSB
- C. CF at LSB
- D. Sign bit at LSB

Answer: B | Logical left shift inserts 0 at the right.

87. The vacant bit inserted by logical SHR is:

- A. 0 at MSB
- B. 1 at MSB
- C. CF at MSB
- D. Old LSB at MSB

Answer: A | Logical right shift inserts 0 at the left.

88. For the shift/rotate count operand in the course slides, the count must be: [TRAP]

- A. Any 8-bit register
- B. Only AL
- C. An immediate value or CL
- D. Only DX

Answer: C | The slide restricts the count to a value or CL.

89. Which is valid according to the shown shift-count rule?

- A. SHL BL,AL
- B. SHL BL,CL
- C. SHL BL,DL
- D. SHL BL,BH

Answer: B | CL is the register used for variable shift counts.

90. How many left shifts are sufficient to force any n-bit register to zero?

- A. n-1
- B. n
- C. 2n
- D. Always 1

Answer: B | After n logical left shifts, all original bits have been shifted out.

91. A one-bit left shift multiplies an unsigned value by 2 only when: [TRAP]

- A. Overflow/wraparound is ignored or does not occur

- B. CF is always 1
- C. The operand is a string
- D. CL is zero

Answer: A | Bit width can discard high bits; the multiplication equivalence assumes no lost significant bit.

92. ROL differs from SHL because ROL:

- A. Discards the MSB
- B. Feeds the shifted-out MSB back into the LSB
- C. Always inserts zero
- D. Does not use CF

Answer: B | ROL circulates the MSB to the LSB and also updates CF.

93. ROR BL,1 places the old LSB into:

- A. Only CF
- B. MSB and CF
- C. LSB and CF
- D. ZF and CF

Answer: B | ROR circulates the old LSB to MSB and copies it to CF.

94. After rotating an n-bit value left exactly n times, the value becomes:

- A. Zero
- B. All ones
- C. The original value
- D. The two's complement

Answer: C | A full-width rotation restores the original bit pattern.

95. Why is rotation convenient for binary output compared with destructive shifting?

- A. It can preserve/restore the original value after a full rotation
- B. It never changes CF
- C. It prints ASCII automatically
- D. It does not use loops

Answer: A | After n rotations the original pattern is restored, unlike shifts that discard bits.

96. RCL rotates through:

- A. Zero Flag
- B. Carry Flag
- C. Stack Pointer
- D. Instruction Pointer

Answer: B | RCL includes CF as an extra bit in the rotation path.

97. The course question about XORing a letter with decimal 32 is based on toggling:

- A. The ASCII case-related bit
- B. All 8 bits
- C. Only the sign flag
- D. The parity flag

Answer: A | Upper/lowercase ASCII letters differ by bit value 32 (20H).

98. Binary keyboard input is not directly a binary integer because the input arrives as:

- A. BCD words
- B. ASCII characters 0 and 1
- C. Hex nibbles only
- D. Raw carry flags

Answer: B | The slide explicitly treats binary input as a character string.

99. In the binary-input algorithm, before inserting each new bit, BX is:

- A. Shifted left
- B. Shifted right
- C. Rotated right
- D. Negated

Answer: A | Left shifting makes room at the LSB for the next bit.

100. The binary-input loop ends when the program reads:

- A. Space
- B. Dollar sign
- C. Carriage return
- D. Zero byte

Answer: C | The algorithm reads until CR.

101. For 16-bit binary output from BX, the slide repeats the output loop:

- A. 4 times
- B. 8 times
- C. 10 times
- D. 16 times

Answer: D | One bit is emitted per rotation, so 16 iterations print a 16-bit word.

102. In binary output using ROL BX, each next bit to print is obtained from:

- A. ZF
- B. CF
- C. PF
- D. AF

Answer: B | ROL moves the current MSB into CF.

103. For hexadecimal input, the register is shifted left by how many bits per input character?

- A. 1
- B. 2
- C. 4
- D. 8

Answer: C | Each hex digit represents 4 bits.

104. The simplified hex-input slide assumes at most:

- A. 2 hex characters
- B. 4 hex characters
- C. 8 hex characters
- D. 16 hex characters

Answer: B | It assumes no more than four uppercase hex characters.

105. For 16-bit hexadecimal output, the algorithm processes:

- A. 2 nibbles
- B. 4 nibbles
- C. 8 nibbles
- D. 16 nibbles

Answer: B | A 16-bit value contains four hexadecimal nibbles.

E. MUL/IMUL, DIV/IDIV & Decimal I/O

106. MUL performs:

- A. Signed multiplication
- B. Unsigned multiplication
- C. Unsigned division
- D. Signed division

Answer: B | MUL treats operands as unsigned.

107. IMUL performs:

- A. Signed multiplication
- B. Unsigned multiplication
- C. Signed division
- D. BCD conversion

Answer: A | IMUL is the signed multiply instruction.

108. For MUL BL, the implicit multiplicand is:

- A. AH
- B. AL
- C. AX
- D. DX

Answer: B | With an 8-bit source, $AX = AL \times \text{source}$.

109. For MUL BX, the implicit multiplicand is:

- A. AL
- B. AX
- C. DX
- D. CX

Answer: B | With a 16-bit source, $DX:AX = AX \times \text{source}$.

110. The product of two 8-bit operands is stored in:

- A. AL only
- B. AX
- C. DX only
- D. DX:AX

Answer: B | An 8x8-bit multiply can produce up to 16 bits, stored in AX.

111. The product of two 16-bit operands is stored in:

- A. AX only
- B. BX:AX
- C. DX:AX
- D. CX:DX

Answer: C | A 16x16-bit product can require 32 bits, stored in DX:AX.

112. After MUL, CF and OF are both 0 when:

- A. The upper half of the product is zero
- B. The lower half is zero
- C. The result is negative

D. ZF is one

Answer: A | For unsigned MUL, upper half zero means the lower half alone holds the result.

113. After IMUL, CF and OF are 0 when the upper half is:

- A. Always zero
- B. A correct sign extension of the lower half
- C. Equal to the divisor
- D. Equal to CF

Answer: B | That means the signed result fits in the lower destination half.

114. Which flags are emphasized as meaningful after MUL/IMUL?

- A. ZF and SF
- B. CF and OF
- C. PF and AF
- D. IF and DF

Answer: B | The slides focus on Carry and Overflow for multiplication.

115. DIV performs:

- A. Signed division
- B. Unsigned division
- C. Unsigned multiplication
- D. Bitwise division

Answer: B | DIV is unsigned division.

116. IDIV performs:

- A. Unsigned division
- B. Signed division
- C. Signed multiplication
- D. Logical right shift

Answer: B | IDIV performs signed division.

117. With an 8-bit divisor, the dividend is:

- A. AL
- B. AX
- C. DX
- D. DX:AX

Answer: B | Byte division divides AX by the 8-bit source.

118. With an 8-bit divisor, the quotient and remainder are stored in:

- A. AH quotient, AL remainder
- B. AL quotient, AH remainder
- C. AX quotient, DX remainder
- D. BL quotient, BH remainder

Answer: B | AL gets quotient and AH gets remainder.

119. With a 16-bit divisor, the dividend is:

- A. AX
- B. BX:AX

C. DX:AX

D. DS:AX

Answer: C | Word division uses the 32-bit dividend DX:AX.

120. With a 16-bit divisor, quotient and remainder go to:

A. AX quotient, DX remainder

B. DX quotient, AX remainder

C. AL quotient, AH remainder

D. BX quotient, CX remainder

Answer: A | AX receives quotient and DX receives remainder.

121. Before unsigned byte DIV when the original dividend is only in AL, the slide says to:

A. CBW

B. CWD

C. MOV AH,0

D. MOV DX,0

Answer: C | Zero-extend AL into AX by clearing AH.

122. Before signed byte IDIV when the dividend starts in AL, use:

A. MOV AH,0

B. CBW

C. CWD

D. XOR DX,DX

Answer: B | CBW sign-extends AL into AX.

123. Before unsigned word DIV when the original dividend is in AX, use:

A. MOV DX,0

B. CBW

C. CWD

D. MOV AH,0

Answer: A | Zero-extend AX into DX:AX by clearing DX.

124. Before signed word IDIV when the dividend is in AX, use:

A. MOV DX,0

B. CBW

C. CWD

D. NEG DX

Answer: C | CWD sign-extends AX into DX:AX.

125. CBW and CWD take:

A. One immediate operand

B. One register operand

C. No explicit operands

D. Two operands

Answer: C | They use fixed registers and specify no operands.

126. Flags after DIV/IDIV are:

A. All guaranteed zero

- B. Defined like CMP
- C. Undefined
- D. Only ZF defined

Answer: C | The Week 11 slide explicitly notes flags are undefined after division.

127. For decimal input, the running numeric total is updated as:

- A. $\text{total} = \text{total} + \text{digit}$
- B. $\text{total} = 2 * \text{total} + \text{digit}$
- C. $\text{total} = 10 * \text{total} + \text{digit}$
- D. $\text{total} = \text{digit} * \text{digit}$

Answer: C | Each decimal digit extends the previous value by one base-10 place.

128. For signed decimal input, the slide tracks a separate sign and applies which instruction at the end for negative input?

- A. NOT
- B. NEG
- C. ROL
- D. TEST

Answer: B | After building the magnitude, NEG is applied when sign is negative.

129. Decimal output extracts digits by repeatedly dividing by:

- A. 2
- B. 8
- C. 10
- D. 16

Answer: C | Division by 10 produces decimal digits as remainders.

130. Why are decimal-output remainders pushed onto the stack? [TRAP]

- A. To sort them
- B. They are generated least-significant first but must print most-significant first
- C. DIV requires stack operands
- D. To set CF

Answer: B | The stack reverses digit order for correct printing.

F. Stack & Procedures

131. The 8086 stack follows:

- A. FIFO
- B. LIFO
- C. Random access only
- D. Round-robin

Answer: B | The slides recap stack behavior as Last-In, First-Out.

132. PUSH SOURCE in the course slide accepts a:

- A. 8-bit byte only
- B. 16-bit register or memory word
- C. 20-bit physical address only
- D. String only

Answer: B | The slide specifies a 16-bit register or memory word.

133. What happens to SP on PUSH?

- A. SP increases by 1
- B. SP decreases by 1
- C. SP decreases by 2
- D. SP increases by 2

Answer: C | 8086 pushes a 16-bit word, so SP is decremented by 2.

134. What happens to SP on POP?

- A. SP decreases by 2
- B. SP increases by 2
- C. SP becomes zero
- D. SP changes by 1

Answer: B | Popping a word advances SP by 2.

135. The top of the stack is addressed using:

- A. CS:IP
- B. DS:SI
- C. SS:SP
- D. ES:DI

Answer: C | SS gives stack segment base and SP gives offset.

136. PUSH changes the source operand itself:

- A. Always
- B. Never; it copies the source
- C. Only for AX
- D. Only if CF=1

Answer: B | The source remains unchanged.

137. POP DEST moves data from:

- A. DS:SI
- B. SS:SP
- C. CS:IP

D. ES:BX

Answer: B | The content at the top of the stack is moved to destination.

138. Which destination is explicitly excluded for POP in the slide?

- A. AX
- B. CX
- C. DX
- D. IP

Answer: D | POP may target a 16-bit register except IP, or a memory word.

139. If AX=3, BX=2 and CX=1 are pushed in that order, the first value popped is:

- A. 3
- B. 2
- C. 1
- D. Undefined

Answer: C | LIFO returns the most recently pushed value first.

140. PUSHF and POPF operate on:

- A. Only AX
- B. Flags
- C. Only CS
- D. Only data variables

Answer: B | They push/pop the flags register state.

141. A procedure in the slides is best described as:

- A. A data segment
- B. A reusable named block of code
- C. A hardware interrupt
- D. A memory model

Answer: B | Procedures are reusable named code blocks called and returned from.

142. If PROC type is omitted, the slide says the procedure is assumed:

- A. FAR
- B. NEAR
- C. INTERRUPT
- D. EXTERN

Answer: B | NEAR is the default type.

143. A NEAR procedure is called from:

- A. A different segment
- B. The same segment
- C. Only ROM
- D. Only the stack segment

Answer: B | NEAR means caller and procedure are in the same segment.

144. A FAR procedure means the caller is:

- A. In a different segment
- B. In the same source line

- C. Always in DS
- D. Not allowed to return

Answer: A | FAR is used when the calling statement is in a different segment.

145. CALL name does which two key things in the slides?

- A. Clears stack and exits
- B. Stores return address on stack and changes IP to procedure
- C. Copies DS to ES and returns
- D. Only sets ZF

Answer: B | CALL saves where to return and transfers control to the procedure.

146. RET returns by:

- A. Popping the return address into IP
- B. Clearing AX
- C. Jumping to MAIN unconditionally
- D. Restoring DS from DX

Answer: A | RET gets the saved return address from the stack into IP.

147. Assembly procedures in the slides have built-in parameter lists like C functions:

- A. Yes, always
- B. No; parameters are arranged by the programmer
- C. Only FAR procedures do
- D. Only MAIN does

Answer: B | Values can be communicated through registers or the stack.

148. A procedure return value in the later array/procedure slide is commonly communicated through:

- A. A register such as AX
- B. The source code comment
- C. Only the flag register
- D. The .MODEL directive

Answer: A | The caller reads the result from an agreed register.

G. Arrays & Addressing Modes

149. The address assigned to the array variable itself is called the array's:

- A. Top address
- B. Base address
- C. Carry address
- D. Return address

Answer: B | The slide calls the address of the array variable the base address.

150. What does 100 DUP(0) mean in GAMMA DW 100 DUP(0)?

- A. One value 100
- B. 100 word elements initialized to zero
- C. 100 uninitialized bytes
- D. Duplicate address 100H

Answer: B | DUP repeats the given initializer the specified number of times.

151. DELTA DB 212 DUP(?) creates:

- A. 212 initialized words
- B. 212 uninitialized bytes
- C. 212 zeroed words
- D. A 212-bit register

Answer: B | DB makes bytes and ? means uninitialized storage.

152. ARRAY DB 5 DUP(?) is analogous to:

- A. Five initialized words
- B. An uninitialized 5-byte array
- C. One byte with value 5
- D. A 5-bit register

Answer: B | It allocates five byte elements without initialization.

153. LEA SI,ARRAY places into SI:

- A. arr[0] value
- B. The base/effective address of ARRAY
- C. The length of ARRAY
- D. The final element

Answer: B | LEA loads the address, not the element value.

154. MOV AL,[SI] means:

- A. Copy SI itself into AL
- B. Load the byte stored at the memory address pointed to by SI
- C. Store AL into SI register
- D. Increment SI

Answer: B | Brackets dereference the address in SI.

155. For a DB array, moving SI to the next element normally needs:

- A. INC SI
- B. ADD SI,2
- C. ADD SI,4

D. DEC SI twice

Answer: A | A byte element occupies one byte.

156. For a DW array, moving SI to the next element normally needs:

- A. INC SI
- B. ADD SI,2
- C. ADD SI,4
- D. SHL SI,4

Answer: B | A word occupies two bytes.

157. During LOOP-based traversal of 5 array elements, CX is typically initialized to:

- A. 0
- B. 1
- C. 5
- D. Array base address

Answer: C | CX holds the number of iterations/elements.

158. In register addressing mode, the operand is:

- A. A constant
- B. A CPU register
- C. A variable address written directly
- D. Always [BX]

Answer: B | Register mode names a register operand.

159. In immediate addressing mode, the operand is:

- A. A constant in the instruction
- B. A memory variable
- C. A pointer register only
- D. The stack top

Answer: A | Immediate mode includes the constant as part of the instruction.

160. In direct addressing mode as taught, the operand is specified by:

- A. A variable/memory location directly
- B. Only a register
- C. The stack pointer only
- D. A literal count only

Answer: A | Direct mode refers directly to a memory variable/address.

161. MOV AX,[SI] is an example of:

- A. Immediate
- B. Register indirect
- C. Direct register
- D. Based indexed with displacement

Answer: B | SI contains the offset address of the memory operand.

162. Which registers are allowed for register-indirect memory addressing in the slide?

- A. AX,CX,DX,SP
- B. BX,SI,DI,BP

- C. CS,DS,SS,ES
- D. AH,AL,BH,BL

Answer: B | The slide lists BX, SI, DI and BP.

163. When BP is used for indirect addressing, its default segment is: [TRAP]

- A. CS
- B. DS
- C. SS
- D. ES

Answer: C | The slide notes BP uses SS, while the others use DS.

164. When SI is used for indirect addressing, its default segment is normally:

- A. CS
- B. DS
- C. SS
- D. ES

Answer: B | SI normally uses DS according to the slide note.

165. An addressing mode using BX or BP plus displacement is called:

- A. Indexed
- B. Based
- C. Immediate
- D. Register only

Answer: B | Using a base register gives based addressing.

166. An addressing mode using SI or DI plus displacement is called:

- A. Based
- B. Indexed
- C. Immediate
- D. Stack

Answer: B | Using an index register gives indexed addressing.

167. [BX+SI] is an example of:

- A. Based indexed addressing
- B. Immediate addressing
- C. Direct addressing
- D. Only register addressing

Answer: A | It combines a base register and an index register.

168. Which instruction loads the value 100H into AX rather than the memory word at DS:0100H when SI=0100H? [TRAP]

- A. MOV AX,[SI]
- B. MOV AX,SI
- C. LEA AX,[SI]
- D. POP AX

Answer: B | Without brackets, MOV AX,SI copies the register value itself.

169. Which mode is especially useful for walking through a one-dimensional array using SI?

- A. Register indirect/indexed style
- B. Immediate only
- C. Flag mode
- D. Segment mode

Answer: A | The additional indirect/indexed modes are used for array processing.

170. Which statement about procedure parameters matches the array/procedure slide?

- A. They must be declared in parentheses
- B. They may be placed in registers or pushed on the stack
- C. Only AX can be used
- D. Procedures cannot receive values

Answer: B | The caller and procedure agree on registers/stack locations.

H1. MDA-8086 Trainer Kit

171. In the MDA-8086 trainer kit, KIT mode takes:

- A. Assembly source directly
- B. Machine language
- C. Only C code
- D. Python bytecode

Answer: B | The Lab 1 sheet says KIT mode takes machine language.

172. PC mode is described as taking:

- A. Only machine code
- B. Assembly or C programming language
- C. Only binary keypad input
- D. Only ROM data

Answer: B | The sheet states PC mode accepts assembly or C programming language.

173. Which keypad key performs a system reset?

- A. RES
- B. REG
- C. MON
- D. STP

Answer: A | RES is labeled system reset.

174. Which keypad key is used to set a memory address?

- A. AD
- B. DA
- C. GO
- D. MON

Answer: A | AD is labeled set memory address.

175. Which keypad key updates segment & offset and inputs data to memory?

- A. DA
- B. REG
- C. RES
- D. STP

Answer: A | DA is the data/memory update key described in the diagram.

176. Which key is described as register display?

- A. REG
- B. GO
- C. MON
- D. AD

Answer: A | REG is labeled Register Display.

177. Which key executes the user program one single step?

- A. GO
- B. STP
- C. RES

D. +

Answer: B | STP is labeled execute user's program, a single step.

178. Which key can go to the user program or execute monitor functions?

- A. MON
- B. GO
- C. REG
- D. DA

Answer: B | GO has that function in the keypad diagram.

179. Which key immediately breaks the user program and is associated with a non-maskable interrupt?

- A. MON
- B. STP
- C. AD
- D. RES

Answer: A | MON is the immediate break / non-maskable interrupt key.

180. The + and - keys are used mainly to:

- A. Change segment & offset/register display up or down
- B. Toggle LCD cursor
- C. Set CF
- D. Start DOS INT 21H

Answer: A | The keypad diagram labels increment/decrement of segment & offset/register display.

H2. LCD Interface with 8086

181. The LCD instruction-register I/O address in the lab manual is:

- A. 00H
- B. 02H
- C. 04H
- D. 1FH

Answer: A | 00H maps to the Instruction Register.

182. The LCD flag/status-register I/O address is:

- A. 00H
- B. 02H
- C. 04H
- D. 18H

Answer: B | 02H maps to the flag/status register.

183. The LCD data-register I/O address is:

- A. 00H
- B. 02H
- C. 04H
- D. 1AH

Answer: C | 04H maps to the Data Register.

184. The LCD controller has how many 8-bit registers emphasized in the manual?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: B | It names the Instruction Register (IR) and Data Register (DR).

185. The Instruction Register (IR) is described as:

- A. Read-only
- B. Write-only
- C. 16-bit
- D. Used only for ASCII data

Answer: B | IR is write-only and stores instruction codes/addresses.

186. The Data Register (DR) stores:

- A. Only flags
- B. Data to be displayed, using character ASCII values
- C. Only cursor addresses
- D. Stack return addresses

Answer: B | DR holds display data; the manual says the data is ASCII value of the character.

187. For a 16x2 LCD, the first line address range is:

- A. 00H-0FH
- B. 40H-4FH
- C. 80H-8FH

D. C0H-CFH

Answer: C | The manual gives 80H through 8FH for line 1.

188. For a 16x2 LCD, the second line address range is:

- A. 10H-1FH
- B. 80H-8FH
- C. C0H-CFH
- D. F0H-FFH

Answer: C | The manual gives C0H through CFH for line 2.

189. In the line address, the second hex digit indicates:

- A. Row number only
- B. Column number 0 through F
- C. Busy state
- D. Interface width

Answer: B | The manual states the second digit indicates the column.

190. RS=0, R/W=0 selects:

- A. IR write / internal command operation
- B. Busy flag read
- C. DR write
- D. DR read

Answer: A | That combination writes the Instruction Register.

191. RS=0, R/W=1 selects:

- A. DR write
- B. Busy flag and address counter read
- C. DR read
- D. Display clear write

Answer: B | This reads BF on DB7 and address counter on DB0-DB6.

192. RS=1, R/W=0 selects:

- A. IR write
- B. Busy read
- C. DR write
- D. DR read

Answer: C | RS=1 selects data register; R/W=0 is write.

193. RS=1, R/W=1 selects:

- A. IR write
- B. Busy read
- C. DR write
- D. DR read

Answer: D | Both high means Data Register read.

194. Where does the LCD busy flag appear?

- A. DB0
- B. DB3

C. DB7

D. RS

Answer: C | BF is output on DB7 during RS=0 read.

195. If the busy flag is 1, the next instruction should:

A. Be written immediately

B. Wait until BF becomes 0

C. Toggle RS twice

D. Clear DR

Answer: B | BF=1 means an internal operation is in progress and new instructions are not accepted.

196. Which instruction takes about 1.64 ms in the table?

A. Clear display

B. Entry mode set

C. Display on/off control

D. Cursor/display shift

Answer: A | Clear display is shown with 1.64 ms execution time.

197. The clear-display command also returns the cursor to:

A. Column F

B. Home position/address 0

C. Second line

D. Last DDRAM address

Answer: B | The table says clear display returns the cursor home.

198. In the LCD bit table, I/D=1 means:

A. Decrement cursor

B. Increment cursor

C. Display off

D. Blink on

Answer: B | I/D selects cursor direction; 1 increments.

199. In the LCD bit table, S=0 means:

A. Shift display

B. No display shift

C. Cursor blink

D. 8-bit interface

Answer: B | S=0 is no display shift.

200. D=1, C=1, B=1 means the respective features are:

A. Display, cursor and blink all on

B. All off

C. Only display on

D. Only blink on

Answer: A | D, C and B are on/off controls for display, cursor and blink.

201. S/C=0 means:

A. Shift display

- B. Move cursor
- C. 8-bit mode
- D. Read DR

Answer: B | S/C=0 selects cursor movement; 1 selects display shift.

202. R/L=1 means:

- A. Shift left
- B. Shift right
- C. No shift
- D. Read status

Answer: B | R/L=1 selects right; 0 selects left.

203. DL=1 means:

- A. 4-bit interface
- B. 8-bit interface
- C. 1-line display
- D. 5x10 dots

Answer: B | DL selects interface width; 1 = 8-bit.

204. N=1 means:

- A. One line duty
- B. Two-line / 1/16 duty
- C. 5x10 font
- D. Busy

Answer: B | N=1 corresponds to 1/16 duty (2 lines).

205. F=1 means:

- A. 5x7 dots
- B. 5x10 dots
- C. Display off
- D. Cursor right

Answer: B | F=1 selects 5x10 dots.

206. BF=0 means:

- A. Internal operation in progress
- B. Can accept an instruction
- C. Display is off
- D. Data register selected

Answer: B | BF=0 indicates the LCD is ready for the next instruction.

H3. 7-Segment Display & Discrete LEDs

207. For the lab 7-segment setup, Port A address is:

- A. 18H
- B. 19H
- C. 1AH
- D. 1BH

Answer: B | Port A is 19H for the 7-segment display.

208. For the four discrete LEDs, Port B address is:

- A. 19H
- B. 1AH
- C. 1BH
- D. 1DH

Answer: C | Port B is 1BH.

209. The 7-segment setup Port C address is:

- A. 1BH
- B. 1CH
- C. 1DH
- D. 1EH

Answer: C | Port C is 1DH.

210. The control-register address for the 7-segment/LED setup is:

- A. 1CH
- B. 1DH
- C. 1EH
- D. 1FH

Answer: D | The control register is 1FH.

211. The 7-segment display in the lab is connected as:

- A. Common cathode
- B. Common anode
- C. Open collector only
- D. Matrix-only

Answer: B | The lab sheet explicitly says common anode.

212. For the common-anode 7-segment setup, a segment glows when its output bit is:

- A. 0
- B. 1
- C. Either always
- D. High impedance only

Answer: A | The sheet states 0 to glow segment LEDs.

213. Which mapping is correct for segment a?

- A. PA7
- B. PA3
- C. PA1

D. PA0

Answer: D | The table maps h g f e d c b a to PA7..PA0, so a=PA0.

214. Which mapping is correct for segment g?

A. PA6

B. PA5

C. PA1

D. PA0

Answer: A | g maps to PA6.

215. Which port drives the four standalone LEDs?

A. Port A

B. Port B

C. Port C

D. LCD DR

Answer: B | The sheet states Port B is used for LED.

216. The standalone LED connection is:

A. Common anode

B. Common cathode

C. No common terminal

D. Series only

Answer: B | The sheet labels the discrete LEDs as common cathode.

217. For the standalone common-cathode LEDs, a bit value of 1 will:

A. Turn the LED on

B. Turn the LED off

C. Reset the CPU

D. Select LCD IR

Answer: A | The sheet says 1 to glow LEDs.

218. Which LED is connected to PB2?

A. Left upper red

B. Green

C. Yellow

D. Bottom lower red

Answer: C | PB2 is the yellow LED.

H4. Dot-Matrix Interface

219. For the dot-matrix setup, Port A address is:

- A. 18H
- B. 19H
- C. 1AH
- D. 1CH

Answer: A | Port A is 18H.

220. For the dot-matrix setup, Port B address is:

- A. 18H
- B. 19H
- C. 1AH
- D. 1BH

Answer: C | Port B is 1AH.

221. For the dot-matrix setup, Port C address is:

- A. 1AH
- B. 1BH
- C. 1CH
- D. 1DH

Answer: C | Port C is 1CH.

222. The dot-matrix control-register address is:

- A. 1CH
- B. 1DH
- C. 1EH
- D. 1FH

Answer: C | Control register is 1EH.

223. In the dot-matrix sheet, Port A is connected to:

- A. Red LEDs
- B. Green LEDs
- C. Columns
- D. LCD data

Answer: B | Port A is connected with green LEDs.

224. In the dot-matrix sheet, Port B is connected to:

- A. Green LEDs
- B. Red LEDs
- C. Columns
- D. 7-segment common pin

Answer: B | Port B is connected with red LEDs.

225. In the dot-matrix sheet, Port C is connected to:

- A. Rows only
- B. Columns
- C. Only red LEDs

D. LCD RS

Answer: B | Port C is connected with columns.

226. Lighting both Port A and Port B LED colors together produces:

- A. Blue
- B. Orange
- C. White
- D. No light

Answer: B | The lab sheet states the combined color is orange.

EXPANDED EDITION - What Was Added

New coverage in this edition

- All 226 questions from the previous master note are preserved.
- 140 additional questions are added (227-366) from the new 35-page MMA quiz note and in-syllabus web patterns.
- New-note focus: MDA-8086 trainer workflow, machine-code memory entry, assembly program formatting, LCD program tracing, busy-flag logic, display blinking, and dot-matrix scanning code.
- Code-trap focus: register/value tracing, flags, shifts/rotates, LOOP, stack order, MUL/DIV implicit registers, arrays, LCD busy polling, and pointer movement.
- WEB-INSPIRED items are paraphrased and strictly filtered to concepts already present in the course materials.

How to solve code MCQs quickly

- Track the exact operand width first: AL/AH are 8-bit; AX/BX/CX/DX/SI are 16-bit.
- After every instruction, write only the registers/flags that actually changed.
- For CMP/TEST, remember that the computed result is used for flags but is not stored.
- For LOOP, the body has already executed before LOOP decrements CX.
- For PUSH/POP, use LIFO and remember SP changes by 2 for a word.
- For MUL/DIV, identify the implicit accumulator/dividend registers before calculating.
- For LCD code, watch AL carefully: ISBUSY reads status into AL, so the character is preserved in AH in the corrected routine.
- For active-low hardware, a 0 can mean ON; do not assume 1 always means ON.

I. NEW NOTE - MDA-8086 Trainer Kit Workflow & PC/KIT Operation

227. According to the newly added trainer-kit note, the RAM size of the trainer board is: [NEW NOTE]

- A. 32 KB
- B. 64 KB
- C. 128 KB
- D. 1 MB

Answer: B | The note explicitly states that the trainer-board RAM size is 64 KB.

228. The trainer-board LCD shown in the new note is described as: [NEW NOTE]

- A. 8 x 1
- B. 16 x 1
- C. 16 x 2
- D. 20 x 4

Answer: C | The note labels the display as a 16 x 2 LCD.

229. In the new note, "standalone" operation means the trainer kit is: [NEW NOTE]

- A. Connected to the PC
- B. Not connected to the PC
- C. Running only C code
- D. Using only the dot matrix

Answer: B | The note defines standalone as the kit not being connected to the PC.

230. In the new note, PC mode means the trainer kit is: [NEW NOTE]

- A. Disconnected from all cables
- B. Connected to the PC
- C. Using only machine code from keypad
- D. Reset after every instruction

Answer: B | PC mode is the kit-connected-to-PC mode in the note.

231. The O/P Select control on the trainer is used mainly to switch between: [NEW NOTE]

- A. LCD and dot matrix
- B. KIT and PC
- C. Read and write
- D. RAM and ROM

Answer: B | The note describes O/P Select as a toggle between KIT and PC sides.

232. The serial connection used for PC mode is identified in the note as the: [NEW NOTE]

- A. PS/2 port
- B. RS-232 port
- C. USB-C port
- D. HDMI port

Answer: B | The note describes the board-to-PC serial link as RS-232.

233. After pressing RESET, the note warns that the user program in memory: [TRAP]

- A. Is always erased
- B. May still remain and should be checked
- C. Is moved to ROM

D. Is automatically executed

Answer: B | RESET restarts the system, but the note says the instructions may still remain in memory.

234. If a user program is currently executing and MON is pressed, the note indicates that control returns to the: [NEW NOTE]

A. User program start

B. Monitor program

C. LCD data register

D. Dot-matrix pattern table

Answer: B | MON breaks the user program and returns the trainer to its monitor program.

235. The AD key is used first when the user wants to: [NEW NOTE]

A. Display a register

B. Enter/set a memory address

C. Execute one instruction

D. Turn off the LCD

Answer: B | AD is the address-entry key.

236. While entering a memory location on the trainer, the colon key is associated with separating: [NEW NOTE]

A. Opcode and operand

B. Segment and offset

C. High byte and low byte

D. Data and parity

Answer: B | The note explains memory addressing in segment:offset form and uses the colon during entry.

237. The data keys used for hexadecimal entry on the trainer cover: [NEW NOTE]

A. 0-7 only

B. 0-9 only

C. 0-F

D. A-Z

Answer: C | Hexadecimal entry uses 0-9 and A-F.

238. After an address has been selected, the DA key is used to: [NEW NOTE]

A. Enter/update data at memory

B. Jump to the monitor

C. Show the stack pointer

D. Reset the trainer

Answer: A | DA is used for data entry/update at the selected memory location.

239. Pressing the + key while viewing memory is intended to move to the: [NEW NOTE]

A. Previous memory location

B. Next memory location

C. Program start only

D. Stack top only

Answer: B | The + key increments to the next address.

240. Pressing the - key while viewing memory is intended to move to the: [NEW NOTE]

A. Next memory location

B. Previous memory location

- C. Next register only
- D. LCD status port

Answer: B | The - key decrements to the previous address.

241. To run a user program from its chosen starting address in KIT mode, the user uses: [NEW NOTE]

- A. GO
- B. REG
- C. RES only
- D. DA

Answer: A | GO starts execution from the specified user-program address.

242. The STP key is especially useful for debugging because it: [NEW NOTE]

- A. Executes one instruction at a time
- B. Clears the whole RAM
- C. Changes KIT mode to PC mode
- D. Displays only the flags

Answer: A | STP performs single-step execution.

243. After one STP step, the Instruction Pointer advances according to: [TRAP]

- A. Always exactly 1 byte
- B. The length of the instruction just executed
- C. Always exactly 2 bytes
- D. The value in CX

Answer: B | The note emphasizes that 8086 instructions have variable lengths, so IP moves by that instruction length.

244. The REG key displays register values in the note primarily in: [NEW NOTE]

- A. Binary only
- B. Decimal only
- C. Hexadecimal
- D. ASCII strings

Answer: C | The register-display values are shown in hexadecimal.

245. In the trainer examples, ORG 1000H indicates that the program is assembled to begin at: [NEW NOTE]

- A. 0000H
- B. 0100H
- C. 1000H
- D. FFFFH

Answer: C | ORG 1000H sets the origin/start location to 1000H.

246. The new note explicitly reminds students that assembly language is: [NEW NOTE]

- A. Case-sensitive
- B. Case-insensitive
- C. Numeric-only
- D. Valid only in uppercase

Answer: B | The note states that assembly language is case insensitive.

J. NEW NOTE - Machine-Code Reading, Program Layout & Assembly Formatting

247. In the teacher-style code format, a label normally identifies: [NEW NOTE]

- A. A named instruction/memory location
- B. Only a decimal constant
- C. Only a comment
- D. Only a flag bit

Answer: A | A label names a code or memory location that can be referenced later.

248. Which symbol is normally written after a label in the teacher-style assembly format? [NEW NOTE]

- A. ;
- B. .
- C. :
- D. #

Answer: C | Labels are written with a colon, for example ABAR:.

249. The opcode field of an assembly instruction tells the processor: [NEW NOTE]

- A. Where the comment begins
- B. Which operation to perform
- C. Which segment is always active
- D. How many pages the PDF has

Answer: B | The opcode is the mnemonic operation such as MOV, ADD, OUT, CMP.

250. Depending on the instruction, the operand field in the taught format may contain: [NEW NOTE]

- A. Only two operands
- B. Only one operand
- C. Zero, one, or two operands
- D. Exactly three operands

Answer: C | The note says instructions may take 0, 1, or 2 operands.

251. A semicolon in the assembly source marks: [NEW NOTE]

- A. A machine opcode
- B. A comment that the processor does not execute
- C. A memory address
- D. A mandatory segment separator

Answer: B | Text after ; is a source-code comment and is not executed.

252. In the KIT-mode example, the machine bytes B0 01 correspond to: [CODE]

- A. MOV AL,01H
- B. MOV AX,0100H
- C. OUT 01H,AL
- D. HLT

Answer: A | The note explicitly identifies B0 01 as MOV AL,01.

253. If B0 01 begins at address 1000H and occupies 2 bytes, the next instruction starts at: [CODE TRAP]

- A. 1001H
- B. 1002H
- C. 1003H
- D. 1010H

Answer: B | A 2-byte instruction at 1000H occupies 1000H and 1001H; the next starts at 1002H.

254. The string stored from address 1017H begins with byte 48H. In the shown "Hello World" message, 48H represents: [NEW NOTE]

- A. H
- B. e
- C. space
- D. W

Answer: A | The first character of the shown message is H, whose ASCII byte in the table is 48H.

255. The "HELLO WORLD" data in the example is followed by which byte to mark the end of the message? [NEW NOTE]

- A. 24H
- B. 0DH
- C. 00H
- D. FFH

Answer: C | The trainer example places a 00H terminator immediately after the message.

256. In the shown machine-code table, the byte F4H at the end corresponds to: [CODE]

- A. INC SI
- B. JMP
- C. HLT
- D. OUT

Answer: C | F4H is the halt instruction used at the end of the sample.

257. Which assembly instruction in the sample is responsible for loading SI with the start address of the message? [CODE]

- A. MOV SI, OFFSET M
- B. MOV AL,[SI]
- C. INC SI
- D. OUT 04H,AL

Answer: A | The assembly version loads the message base address into SI.

258. In the sample program, MOV AL,[SI] means: [CODE TRAP]

- A. Copy SI itself into AL
- B. Read the byte from memory pointed to by SI into AL
- C. Store AL into SI
- D. Add SI to AL

Answer: B | Square brackets mean dereference the memory location addressed by SI.

259. In the sample message loop, CMP AL,00H is used to test whether: [CODE]

- A. The LCD is busy
- B. The end-of-message marker has been reached

C. CX is zero

D. SI is even

Answer: B | 00H is the message terminator in this code.

260. The instruction MOV CX,3000H in the sample is used mainly to prepare: [CODE]

A. A delay loop

B. A stack frame

C. A segment address

D. A 7-segment pattern

Answer: A | CX is loaded with a count and then LOOP is used to create a delay.

261. What is the purpose of LOOP L in the sample delay block? [CODE]

A. Increment SI

B. Decrement CX and repeat until CX becomes zero

C. Set CF repeatedly

D. Write to the LCD port

Answer: B | LOOP automatically decrements CX and branches while CX is not zero.

262. OUT 04H,AL in the sample sends AL to the: [CODE]

A. LCD instruction register

B. LCD status register

C. LCD data register

D. Dot-matrix control register

Answer: C | 04H is the LCD Data Register port.

263. The purpose of INC SI inside the message-print loop is to: [CODE]

A. Move to the next character address

B. Increase the LCD cursor register directly

C. Increase CX

D. Set the busy flag

Answer: A | SI is the pointer to the message, so incrementing it advances to the next byte.

264. The backward JMP in the message routine is used so that the program: [CODE]

A. Prints only one character

B. Returns to fetch/check the next character

C. Resets the trainer

D. Changes to PC mode

Answer: B | The unconditional jump returns control to the character-processing loop.

265. Why can the address of instructions after the message change when the message length changes? [TRAP]

A. Because 8086 uses only fixed 8-byte instructions

B. Because the following bytes must be placed immediately after the data block

C. Because the LCD changes CS

D. Because REG rewrites memory

Answer: B | The note points out that changing the number of message bytes shifts the locations of later bytes/instructions.

266. The source-file extension used for the assembly program in PC mode is: [NEW NOTE]

A. .exe

B. .asm

C. .txt

D. .bin only

Answer: B | The note instructs students to create an assembly source file with the .asm extension.

K. NEW NOTE - LCD Program Tracing, Busy Flag, Cursor Position & Blinking

267. Which value is sent to the LCD Instruction Register to clear the display in the new code? [NEW NOTE]

- A. 00H
- B. 01H
- C. 04H
- D. 0FH

Answer: B | The clear-display command is 01H.

268. What does MOV AL,1 followed by OUT 00H,AL do in the LCD program? [CODE]

- A. Reads the busy flag
- B. Clears the display
- C. Writes character 01H to the screen
- D. Turns on red dot-matrix LEDs

Answer: B | 00H is the Instruction Register port and 01H is the clear-display command.

269. The LCD code uses C4H as a cursor address. This places the cursor on: [TRAP]

- A. First line, column 4
- B. Second line, column 4
- C. Second line, column 12
- D. First line, column 12

Answer: B | Second-line addresses begin at C0H; C4H is column 4 when columns are counted from 0.

270. Why does C4H select second-line column 4? [NEW NOTE]

- A. C0H is the second-line base and 4 is the column offset
- B. C4H is the busy flag
- C. C means cursor and 4 means data register
- D. C4H is always the first LCD cell

Answer: A | The lab note maps C0H-CFH to the second line and uses the low hex digit as the column.

271. The value 0CH (00001100B) used as a display-control command corresponds to: [CODE TRAP]

- A. Display on, cursor off, blink off
- B. Display off, cursor on, blink on
- C. Display on, cursor on, blink on
- D. Clear display

Answer: A | For 00001DCB, 0CH gives D=1, C=0, B=0.

272. MOV SI, OFFSET M is used so that SI contains: [CODE]

- A. The ASCII value of M
- B. The starting address of the message
- C. The LCD status value
- D. The current CX count

Answer: B | OFFSET M is the address of the message data.

273. In M DB "HELLO WORLD",00, the final 00H is used as: [NEW NOTE]

- A. A newline
- B. A string terminator checked by the program
- C. A space
- D. The LCD clear command

Answer: B | The code checks for 00H to know that the message is finished.

274. In the corrected LCD loop, MOV AH,[SI] loads: [CODE]

- A. The current message character
- B. The LCD busy flag
- C. The segment address
- D. The loop counter

Answer: A | SI points at the current message byte, which is copied into AH.

275. CMP AH,0 followed by JE FINISH causes a jump when: [CODE]

- A. The LCD is busy
- B. The current message byte is the 00H terminator
- C. The display is on
- D. SI equals zero

Answer: B | JE is taken because CMP sets equality when AH is 0.

276. Why does the corrected version keep the current character in AH rather than AL while calling ISBUSY? [TRAP]

- A. ISBUSY uses and changes AL
- B. AH is the only register that OUT accepts
- C. AL cannot hold ASCII
- D. AH is the stack pointer

Answer: A | ISBUSY reads status into AL, so keeping the character in AH prevents it from being overwritten.

277. After CALL ISBUSY, why does the program execute MOV AL,AH before OUT 04H,AL? [CODE TRAP]

- A. To restore the saved character into the output register
- B. To clear the busy flag manually
- C. To change the cursor address
- D. To reset SI

Answer: A | The character is preserved in AH while ISBUSY uses AL; it is copied back before writing the Data Register.

278. OUT 04H,AL in the LCD character loop writes AL to the: [CODE]

- A. Instruction Register
- B. Flag/Status Register
- C. Data Register
- D. 8255 control register

Answer: C | 04H is the LCD Data Register port used for printable character data.

279. Immediately after a character is written, INC SI prepares the program to: [CODE]

- A. Check the next character
- B. Read the status register twice
- C. Move the cursor home
- D. Reset CX

Answer: A | SI advances to the next message byte.

280. JMP ABAR at the end of the character path makes the routine: [CODE]

- A. Terminate immediately
- B. Repeat character processing
- C. Enter the monitor program
- D. Clear the display every time

Answer: B | ABAR is the top of the message-processing loop.

281. In the non-blinking version, FINISH: HLT means the processor: [CODE]

- A. Continues printing spaces
- B. Stops program execution at that point
- C. Turns on the LCD blink bit
- D. Returns from a subroutine

Answer: B | HLT is used as the final stop after the message is complete.

282. The first instruction of ISBUSY is IN AL,2. It reads from: [CODE]

- A. LCD Instruction Register
- B. LCD Flag/Status Register
- C. LCD Data Register
- D. Dot-matrix Port A

Answer: B | Port 02H is the LCD Flag/Status Register.

283. AND AL,10000000B in ISBUSY is used to isolate: [CODE TRAP]

- A. DB0
- B. DB3
- C. DB7 busy flag
- D. All lower 7 address bits only

Answer: C | The mask keeps only bit 7, where the busy flag resides.

284. If the LCD status byte is 80H, what is the result of AND AL,80H? [CODE]

- A. 00H
- B. 01H
- C. 7FH
- D. 80H

Answer: D | $80H \text{ AND } 80H = 80H$, so the busy bit is still set.

285. After AND AL,80H produces 80H, JNZ ISBUSY will: [CODE]

- A. Fall through to RET
- B. Jump back and poll again
- C. Clear the display
- D. Increment SI

Answer: B | The result is nonzero, meaning $DB7=1$ and the LCD is still busy.

286. If the LCD status byte is 04H, then after AND AL,80H the result is: [CODE]

- A. 00H
- B. 04H
- C. 80H
- D. 84H

Answer: A | Bit 7 is not set in 04H, so masking with 80H produces zero.

287. When the masked busy result becomes zero, ISBUSY executes: [CODE]

- A. JMP ISBUSY forever
- B. RET to the caller
- C. HLT
- D. OUT 04H,AL

Answer: B | ZF=1 makes JNZ not taken, so the subroutine returns.

288. The reason an LCD-busy check is needed before the next command/data write is that the LCD:
[NEW NOTE]

- A. Cannot accept a new write while an internal operation is still in progress
- B. Forgets the cursor address immediately
- C. Uses only decimal input
- D. Always clears DB7 by software

Answer: A | The lab manual states that BF=1 means an internal operation is in progress and the next instruction is not accepted.

289. In the blinking version, when the message terminator is detected, the code jumps to: [CODE]

- A. FINISH
- B. BB
- C. ISBUSY
- D. ABAR

Answer: B | The blinking edition changes the end-of-message branch to BB.

290. Which sequence correctly describes the BB blinking loop? [CODE]

- A. DISON -> WAIT -> HLT
- B. WAIT -> DISOFF -> WAIT -> DISON -> repeat
- C. ISBUSY -> INC SI -> RET
- D. CLEAR -> OUT DATA -> FINISH

Answer: B | The note shows two waits surrounding display off/on, followed by JMP BB.

291. The first CALL WAIT in BB mainly provides: [CODE]

- A. A visible delay before the next display-state change
- B. A memory address conversion
- C. A busy-flag mask
- D. A stack reset

Answer: A | WAIT is a software delay loop so the on/off states are visible.

292. The DISOFF subroutine loads AL with: [NEW NOTE]

- A. 01H
- B. 08H
- C. 0CH
- D. 0FH

Answer: B | The new note uses 08H for display off.

293. The DISON subroutine loads AL with: [NEW NOTE]

- A. 08H
- B. 0CH
- C. 0FH
- D. 80H

Answer: C | The blinking code uses 0FH for display on with cursor and blink enabled.

294. For the Display On/Off format 00001DCB, 0FH gives: [TRAP]

- A. D=0,C=0,B=0
- B. D=1,C=1,B=1
- C. D=1,C=0,B=0
- D. D=0,C=1,B=1

Answer: B | 0FH = 00001111B, so D, C and B are all 1.

295. For the same 00001DCB format, 08H gives: [TRAP]

- A. D=0,C=0,B=0
- B. D=1,C=1,B=1
- C. D=1,C=0,B=0
- D. D=0,C=1,B=0

Answer: A | 08H = 00001000B, so the D/C/B control bits are all 0 and the display is off.

296. Before DISON or DISOFF writes its command, the subroutine calls ISBUSY mainly to: [CODE]

- A. Ensure the LCD is ready to accept the command
- B. Load SI with the message
- C. Reset CX
- D. Change PC mode

Answer: A | The display-control command must be sent only when the LCD is free.

297. OUT 00H,AL inside DISON/DISOFF targets the LCD: [CODE]

- A. Instruction Register
- B. Status Register
- C. Data Register
- D. Dot-matrix Port C

Answer: A | Display on/off is an LCD command and therefore goes to the Instruction Register at 00H.

298. The WAIT subroutine starts with MOV CX,5000H. CX is used as: [CODE]

- A. The delay-loop counter
- B. The message pointer
- C. The LCD data byte
- D. The segment register

Answer: A | LOOP uses CX as its counter.

299. In WAIT, the instruction LOOP L repeatedly: [CODE]

- A. Increments CX until FFFFH
- B. Decrements CX and jumps while it is nonzero
- C. Changes AL between 08H and 0FH
- D. Reads DB7

Answer: B | LOOP decrements CX automatically and repeats until CX reaches zero.

300. After the WAIT loop finishes, RET: [CODE]

- A. Returns to the caller
- B. Restarts the whole program at 1000H
- C. Clears the message
- D. Sets the busy bit

Answer: A | RET returns from the delay subroutine.

301. Why does the blinking program continue indefinitely after printing "HELLO WORLD"? [CODE TRAP]

- A. BB ends with JMP BB
- B. HLT automatically restarts it
- C. ISBUSY sets CX to FFFFH
- D. SI is never incremented

Answer: A | The BB block unconditionally jumps back to itself after each off/on cycle.

L. NEW NOTE - Dot-Matrix Program Tracing

302. In the dot-matrix program, MOV AL,10000000B followed by OUT PCR,AL is used to: [CODE]

- A. Send the 8255 control word
- B. Print a character on LCD
- C. Load the message pointer
- D. Clear CX

Answer: A | The code labels this as sending the control word to the control register.

303. The dot-matrix code initializes AH to 00000001B mainly to hold: [CODE]

- A. A moving column-select bit pattern
- B. The LCD busy flag
- C. The message terminator
- D. The stack top

Answer: A | AH is rotated so a single 1 moves across the column-selection output.

304. Why does the code initially send 1111111B to Port B? [TRAP]

- A. To keep the red row LEDs inactive in the active-low row scheme
- B. To turn on all red LEDs
- C. To select every column
- D. To clear the LCD

Answer: A | The sheet shows row LEDs glowing on 0; all ones therefore keep Port B red LEDs off.

305. MOV SI,OFFSET PAT prepares SI to point to: [CODE]

- A. The eight-byte dot-matrix pattern table
- B. The LCD status register
- C. The current stack frame
- D. The trainer monitor ROM

Answer: A | PAT is the pattern-data table at the end of the program.

306. Inside ABAR, MOV AL,AH followed by OUT PC,AL sends the current: [CODE]

- A. Column-select pattern
- B. Green row pattern
- C. Red row pattern
- D. LCD cursor command

Answer: A | Port C is connected to the columns, and AH carries the rotating selection bit.

307. MOV AL,[SI] in the dot-matrix loop fetches: [CODE]

- A. The current pattern byte from memory
- B. The SI register number
- C. The current column address from Port C
- D. The busy flag

Answer: A | SI points into PAT, so [SI] is the current pattern-data byte.

308. OUT PA,AL sends the fetched byte to: [CODE]

- A. Green LED rows
- B. Red LED rows
- C. LCD data register

D. Trainer keypad

Answer: A | Port A is connected to the green LEDs.

309. MOV CX,30H followed by LOOP L1 is included primarily to: [CODE]

- A. Create a short delay
- B. Compute a physical address
- C. Set the carry flag permanently
- D. Read the next message byte

Answer: A | The empty counted loop delays column scanning.

310. After the delay, INC SI causes the next iteration to use: [CODE]

- A. The next byte in PAT
- B. The previous byte in PAT
- C. The same byte forever
- D. The LCD command at 00H

Answer: A | SI advances by one because PAT is a DB byte array.

311. ROL AH,1 in the dot-matrix program mainly: [CODE]

- A. Moves the active column-selection bit to the next position
- B. Clears every column
- C. Divides AH by two
- D. Loads the next pattern byte

Answer: A | Rotating the one-hot bit scans across columns.

312. JNC ABAR after ROL AH,1 tests whether: [CODE TRAP]

- A. The rotation has wrapped a 1 through the carry flag
- B. SI has reached zero
- C. The LCD is busy
- D. CX is negative

Answer: A | CF receives the rotated-out MSB; JNC repeats while no wrap carry is produced.

313. How many data bytes are defined in PAT in the shown dot-matrix code? [NEW NOTE]

- A. 4
- B. 6
- C. 8
- D. 16

Answer: C | The PAT declaration contains eight DB values, matching the eight matrix columns.

314. Because PAT is declared with DB, advancing to the next element requires SI to change by: [CODE]

- A. 1 byte
- B. 2 bytes
- C. 4 bytes
- D. 8 bytes

Answer: A | DB elements are one byte each, so INC SI advances one pattern element.

315. If a Port A pattern byte is FFH, the active-low row rule implies the green row LEDs for that byte are: [TRAP]

- A. All inactive
- B. All active

C. Only the MSB active

D. Alternating

Answer: A | For the row side, 0 is the glow level; FFH contains no zeros.

316. The dot-matrix code uses Port B=FFH while driving patterns through Port A. This makes the displayed pattern primarily: [CODE TRAP]

A. Green

B. Red

C. Orange

D. Blue

Answer: A | Port A drives green, while all-ones on the active-low red rows keep Port B red LEDs off.

M. WEB-INSPIRED - In-Syllabus 8086 MCQ Patterns

317. Which instruction is the clearest example of immediate addressing? [WEB-INSPIRED]

- A. MOV AX,1234H
- B. MOV AX,[1234H]
- C. MOV AX,BX
- D. MOV AX,[BX]

Answer: A | The constant 1234H is embedded directly in the instruction. This pattern is common in 8086 addressing-mode MCQs.

318. Which instruction uses direct memory addressing? [WEB-INSPIRED]

- A. MOV AX,1234H
- B. MOV AX,[2500H]
- C. MOV AX,BX
- D. MOV AX,[SI]

Answer: B | The effective memory address 2500H is written directly in the instruction.

319. Which instruction uses register-indirect addressing? [WEB-INSPIRED]

- A. MOV AX,[BX]
- B. MOV AX,BX
- C. MOV AX,1234H
- D. MOV AX,[2500H]

Answer: A | BX contains the effective address of the memory operand.

320. Which instruction is an example of based-indexed addressing? [WEB-INSPIRED]

- A. MOV AX,[BX+SI]
- B. MOV AX,[2500H]
- C. MOV AX,SI
- D. MOV AX,5

Answer: A | A base register and an index register are combined to form the effective address.

321. Which instruction uses an index register to access memory with a displacement? [WEB-INSPIRED]

- A. MOV AL,[SI+2]
- B. MOV AL,2
- C. MOV AL,SI
- D. MOV AL,[BX]

Answer: A | SI is the index register and 2 is the displacement.

322. After TEST AL,AL, ZF becomes 1 exactly when: [WEB-INSPIRED]

- A. AL is zero
- B. AL is odd
- C. AL has MSB=1
- D. CF was previously 1

Answer: A | TEST AL,AL forms AL AND AL only for flags; the result is zero only if AL itself is zero.

323. Which instruction pair is specifically used to enter a procedure and then return? [WEB-INSPIRED]

- A. CALL and RET
- B. JMP and LOOP
- C. PUSH and POP only
- D. CMP and JE

Answer: A | CALL transfers control to a procedure and RET returns to the saved return address.

324. Given PUSH AX followed by PUSH BX, which value is obtained by the first POP? [WEB-INSPIRED]

- A. The old AX value
- B. The old BX value
- C. The old CX value
- D. Always zero

Answer: B | The stack is LIFO, so the most recently pushed BX value is popped first.

325. For an unsigned value, SHL AL,1 approximately performs which arithmetic operation when no significant bit is lost? [WEB-INSPIRED]

- A. $AL \times 2$
- B. $AL / 2$
- C. $AL + 1$
- D. $-AL$

Answer: A | A one-bit left shift multiplies an unsigned value by 2 if overflow is ignored.

326. For an unsigned value, SHR AL,1 approximately performs: [WEB-INSPIRED]

- A. $AL \times 2$
- B. $AL / 2$ with truncation
- C. $AL + 2$
- D. Two's complement

Answer: B | Logical right shift divides an unsigned value by 2, discarding the remainder bit.

327. For an 8-bit operand, ROL AL,8 returns the bit pattern to: [WEB-INSPIRED]

- A. All zeros
- B. All ones
- C. Its original arrangement
- D. The two's complement

Answer: C | Eight one-bit rotations cycle every bit through all positions and back to its start.

328. Which instruction is a common one-instruction way to clear a register without using an immediate zero? [WEB-INSPIRED]

- A. XOR AL,AL
- B. OR AL,AL
- C. TEST AL,AL
- D. NOT AL

Answer: A | A value XOR itself is zero.

329. AND AL,0FH is commonly used to: [WEB-INSPIRED]

- A. Keep only the low nibble
- B. Keep only the high nibble
- C. Set every bit
- D. Rotate four places

Answer: A | 0FH has ones only in the low four bit positions.

330. OR AL,80H guarantees that: [WEB-INSPIRED]

- A. The MSB becomes 1
- B. The LSB becomes 0
- C. AL becomes zero
- D. CF is cleared

Answer: A | OR with 10000000B forces bit 7 to one and leaves other bits unchanged.

331. TEST AL,01H is a compact way to check whether an integer in AL is: [WEB-INSPIRED]

- A. Odd or even
- B. Positive or negative only
- C. Greater than 255
- D. A valid ASCII letter

Answer: A | Bit 0 determines parity of an integer: 1 for odd, 0 for even.

N. TRICKY ASSEMBLY CODE MCQs - Trace Before You Answer

332. What is the final value of AL? [CODE TRAP]

```
MOV AL,05H  
ADD AL,03H
```

- A. 08H
- B. 05H
- C. 03H
- D. 00H

Answer: A | $5 + 3 = 8$.

333. What is the final 8-bit hexadecimal value of AL? [CODE TRAP]

```
MOV AL,05H  
SUB AL,07H
```

- A. 02H
- B. FEH
- C. FCH
- D. 07H

Answer: B | $05H - 07H = -2$, represented as FEH in 8-bit two's complement.

334. What is the final value of AL? [CODE TRAP]

```
MOV AL,0FH  
AND AL,33H
```

- A. 03H
- B. 0FH
- C. 30H
- D. 3FH

Answer: A | $0FH \text{ AND } 33H = 03H$.

335. What is the final value of AL? [CODE TRAP]

```
MOV AL,55H  
OR AL,0FH
```

- A. 50H
- B. 55H
- C. 5FH
- D. FFH

Answer: C | $55H \text{ OR } 0FH$ sets the low nibble to F, giving 5FH.

336. What is the final value of AL? [CODE TRAP]

```
MOV AL,0AAH  
XOR AL,0FFH
```

- A. 00H
- B. 55H
- C. AAH
- D. FFH

Answer: B | $AAH \text{ XOR } FFH$ complements every bit, producing 55H.

337. After the code executes, which pair is correct? [CODE TRAP]

```
MOV AL,81H
SHL AL,1
```

- A. AL=02H, CF=1
- B. AL=02H, CF=0
- C. AL=81H, CF=1
- D. AL=40H, CF=1

Answer: A | SHL discards the old MSB into CF and inserts 0 at the LSB. 81H becomes 02H with CF=1.

338. After the code executes, which pair is correct? [CODE TRAP]

```
MOV AL,81H
SHR AL,1
```

- A. AL=40H, CF=1
- B. AL=C0H, CF=1
- C. AL=02H, CF=0
- D. AL=80H, CF=0

Answer: A | SHR moves the old LSB to CF and shifts in 0 at the MSB. 81H becomes 40H and CF=1.

339. After the code executes, which pair is correct? [CODE TRAP]

```
MOV AL,81H
ROL AL,1
```

- A. AL=03H, CF=1
- B. AL=02H, CF=1
- C. AL=C0H, CF=1
- D. AL=81H, CF=0

Answer: A | ROL moves the old MSB into both CF and the LSB, so 81H becomes 03H with CF=1.

340. After the code executes, which pair is correct? [CODE TRAP]

```
MOV AL,81H
ROR AL,1
```

- A. AL=40H, CF=1
- B. AL=C0H, CF=1
- C. AL=03H, CF=1
- D. AL=80H, CF=0

Answer: B | ROR moves the old LSB to CF and back into the MSB, giving C0H with CF=1.

341. What is the final value of AL? [CODE TRAP]

```
MOV AL,01H
NEG AL
```

- A. 01H
- B. 7FH
- C. FFH
- D. 00H

Answer: C | NEG computes two's complement arithmetic negation; -1 is FFH in 8 bits.

342. What is the final value of AL? [CODE TRAP]

```
MOV AL,0FFH
NOT AL
```

- A. 00H
- B. 01H
- C. 7FH
- D. FFH

Answer: A | NOT flips every bit, so FFH becomes 00H.

343. After TEST, what is the value of ZF? [CODE TRAP]

```
MOV AL,04H
TEST AL,01H
```

- A. ZF=0
- B. ZF=1
- C. ZF is always undefined
- D. ZF equals CF

Answer: B | 04H AND 01H is zero, so TEST sets ZF=1.

344. After TEST, what is the value of ZF? [CODE TRAP]

```
MOV AL,05H
TEST AL,01H
```

- A. ZF=0
- B. ZF=1
- C. ZF is unchanged by TEST
- D. ZF becomes the LSB

Answer: A | 05H AND 01H is 1, a nonzero result, so ZF=0.

345. What is AL after the loop finishes? [CODE TRAP]

```
MOV AL,00H
MOV CX,3
L1: INC AL
    LOOP L1
```

- A. 00H
- B. 01H
- C. 02H
- D. 03H

Answer: D | The loop body executes three times because CX starts at 3.

346. Is the jump taken? [CODE TRAP]

```
MOV CX,0
JCXZ DONE
```

- A. Yes, because CX is zero
- B. No, because ZF is unknown
- C. Only if CF=1
- D. Only after LOOP executes

Answer: A | JCXZ checks CX directly and jumps when CX=0.

347. Which statement is correct after CMP? [CODE TRAP]

```
MOV AX,1234H
MOV BX,1234H
CMP AX,BX
JE SAME
```

- A. AX becomes 0
- B. BX becomes 0
- C. AX and BX remain unchanged and JE is taken
- D. AX and BX are swapped

Answer: C | CMP updates flags as if subtraction occurred but does not store the result; equal operands make JE true.

348. What values are placed in CX and DX? [CODE TRAP]

```
MOV AX,1111H
MOV BX,2222H
PUSH AX
PUSH BX
POP CX
POP DX
```

- A. CX=1111H, DX=2222H
- B. CX=2222H, DX=1111H
- C. CX=2222H, DX=2222H
- D. CX=1111H, DX=1111H

Answer: B | LIFO order pops the last pushed BX value first, then the earlier AX value.

349. If SP starts at 1000H, what are SP values immediately after PUSH AX and then after POP BX? [CODE TRAP]

```
PUSH AX
POP BX
```

- A. 0FFFH then 1000H
- B. 0FFEh then 1000H
- C. 1002H then 1000H
- D. 0FFEh then 0FFCH

Answer: B | A word PUSH decrements SP by 2; POP restores it by adding 2.

350. What is AX after MUL BL? [CODE TRAP]

```
MOV AL,0AH
MOV BL,03H
MUL BL
```

- A. 000DH
- B. 001EH
- C. 030AH
- D. 1E00H

Answer: B | Byte MUL uses AL as the implicit operand: 0AH x 03H = 1EH, stored in AX.

351. After MUL BX, what is DX:AX? [CODE TRAP]

```
MOV AX,0100H
MOV BX,0010H
MUL BX
```

- A. 0000H:1000H
- B. 1000H:0000H
- C. 0001H:0000H
- D. 0000H:0100H

Answer: A | 0100H x 0010H = 1000H, which fits in AX, so DX=0000H.

352. After DIV BL, what are AL and AH? [CODE TRAP]

```
MOV AX,0011H
MOV BL,05H
DIV BL
```

- A. AL=2, AH=3
- B. AL=3, AH=2
- C. AL=5, AH=2
- D. AL=17, AH=0

Answer: B | 17 / 5 gives quotient 3 in AL and remainder 2 in AH for byte division.

353. After DIV BX, what are AX and DX? [CODE TRAP]

```
MOV DX,0
MOV AX,0011H
MOV BX,0005H
DIV BX
```

- A. AX=2, DX=3
- B. AX=3, DX=2
- C. AX=5, DX=2
- D. AX=17, DX=0

Answer: B | For word DIV, DX:AX is divided by BX; quotient goes to AX and remainder to DX.

354. What is the purpose of CBW in this signed example? [CODE TRAP]

```
MOV AL,-5
CBW
```

- A. Clear BX
- B. Sign-extend AL into AX
- C. Clear the carry flag
- D. Convert binary to ASCII

Answer: B | CBW copies the sign of AL through AH, preparing an 8-bit signed value as a 16-bit signed value.

355. Why is DX cleared before an unsigned word DIV? [CODE TRAP]

```
MOV DX,0
DIV BX
```

- A. To make the high half of the dividend zero
- B. To reset the stack
- C. To make BX positive
- D. To clear the Zero Flag

Answer: A | Word DIV uses DX:AX as the dividend; for an unsigned AX-only value, DX should be zero.

356. If BX=0005H before these two instructions, what is BX afterward? [CODE TRAP]

```
SHL BX,1
OR BX,1
```

- A. 0005H
- B. 000AH
- C. 000BH
- D. 0011H

Answer: C | SHL doubles 5 to 10, then OR with 1 inserts the new low input bit, producing 11 decimal = 000BH.

357. After one ROL BL,1, where can the bit that left the MSB be tested? [CODE TRAP]

```
ROL BL,1
```

- A. ZF only
- B. CF
- C. SP
- D. DS

Answer: B | ROL copies the old MSB into CF while also rotating it into the LSB.

358. What value is loaded into AL? [CODE TRAP]

```
ARR DB 10,20,30
LEA SI,ARR
INC SI
MOV AL,[SI]
```

- A. 10
- B. 20
- C. 30
- D. The address of ARR

Answer: B | SI first points at ARR[0]=10; INC SI advances one byte to ARR[1]=20.

359. What value is loaded into AX? [CODE TRAP]

```
W DW 10,20,30
LEA SI,W
ADD SI,2
MOV AX,[SI]
```

- A. 10
- B. 20
- C. 30
- D. The address of W

Answer: B | DW elements are 2 bytes, so ADD SI,2 advances from W[0] to W[1]=20.

360. How many times does BODY execute? [CODE TRAP]

```
MOV CX,5
L1: ; BODY
LOOP L1
```

- A. 4
- B. 5
- C. 6
- D. Until ZF=1

Answer: B | LOOP starts with CX=5 and decrements after each iteration, so the body runs five times.

361. How many times does BODY execute? [CODE TRAP]

```
MOV CX,1
L1: ; BODY
LOOP L1
```

- A. 0
- B. 1
- C. 2
- D. 65536

Answer: B | The body runs once before LOOP decrements CX from 1 to 0 and stops.

362. If the LCD status value read is 80H, what happens at JNZ? [CODE TRAP]

```
IN AL,2
AND AL,80H
JNZ ISBUSY
```

- A. The jump is taken back to ISBUSY
- B. The jump is not taken
- C. The display is cleared
- D. SI increments

Answer: A | 80H AND 80H is nonzero, so JNZ repeats the busy poll.

363. If the LCD status value read is 04H, what happens after the mask? [CODE TRAP]

```
IN AL,2
AND AL,80H
JNZ ISBUSY
RET
```

- A. JNZ is taken
- B. JNZ is not taken and RET can execute
- C. AL becomes 84H
- D. The LCD is necessarily busy

Answer: B | 04H AND 80H = 00H, so ZF=1 and JNZ is not taken.

364. Suppose AH contains ASCII "A" before CALL ISBUSY. What will OUT 04H,AL send after MOV AL,AH? [CODE TRAP]

```
MOV AH,'A'
CALL ISBUSY
MOV AL,AH
OUT 04H,AL
```

- A. The busy status byte
- B. ASCII "A"
- C. The value of SI
- D. Always 00H

Answer: B | AH preserves the character while ISBUSY uses AL; MOV AL,AH restores "A" for output.

365. What character is in AH after the shown sequence? [CODE TRAP]

```
M DB 'HI',00
MOV SI,OFFSET M
MOV AH,[SI]
INC SI
MOV AH,[SI]
```

- A. H
- B. I
- C. 00H
- D. The address of M

Answer: B | SI starts at H, is incremented once, then [SI] is the second character I.

366. In the blinking LCD program, what happens after the message terminator is detected? [CODE TRAP]

```
CMP AH,0
JE BB
```

```
...  
BB: CALL WAIT  
    CALL DISOFF  
    CALL WAIT  
    CALL DISON  
    JMP BB
```

- A. The program halts immediately
- B. Control enters an endless WAIT/DISOFF/WAIT/DISON loop
- C. SI is reset and the message reprints
- D. The trainer automatically enters PC mode

Answer: B | The terminator branch targets BB, and BB ends with JMP BB, so the display keeps blinking.

Rapid Answer Key - 1 to 366

1-B 2-C 3-C 4-B 5-B 6-A 7-C 8-C 9-C 10-A 11-B 12-D 13-A 14-A 15-B 16-B 17-A 18-B 19-C 20-C
 21-B 22-C 23-B 24-B 25-B 26-B 27-B 28-C 29-D 30-B 31-B 32-B 33-B 34-B 35-B 36-C 37-C 38-C 39-C 40-B
 41-A 42-B 43-A 44-A 45-B 46-C 47-C 48-C 49-B 50-B 51-B 52-B 53-B 54-A 55-A 56-B 57-C 58-D 59-C 60-B
 61-A 62-B 63-B 64-C 65-B 66-B 67-B 68-B 69-B 70-A 71-A 72-B 73-B 74-C 75-B 76-B 77-B 78-B 79-A 80-B
 81-B 82-B 83-B 84-B 85-B 86-B 87-A 88-C 89-B 90-B 91-A 92-B 93-B 94-C 95-A 96-B 97-A 98-B 99-A 100-C
 101-D 102-B 103-C 104-B 105-B 106-B 107-A 108-B 109-B 110-B 111-C 112-A 113-B 114-B 115-B 116-B 117-B 118-B 119-C 120-A
 121-C 122-B 123-A 124-C 125-C 126-C 127-C 128-B 129-C 130-B 131-B 132-B 133-C 134-B 135-C 136-B 137-B 138-D 139-C 140-B
 141-B 142-B 143-B 144-A 145-B 146-A 147-B 148-A 149-B 150-B 151-B 152-B 153-B 154-B 155-A 156-B 157-C 158-B 159-A 160-A
 161-B 162-B 163-C 164-B 165-B 166-B 167-A 168-B 169-A 170-B 171-B 172-B 173-A 174-A 175-A 176-A 177-B 178-B 179-A 180-A
 181-A 182-B 183-C 184-B 185-B 186-B 187-C 188-C 189-B 190-A 191-B 192-C 193-D 194-C 195-B 196-A 197-B 198-B 199-B 200-A
 201-B 202-B 203-B 204-B 205-B 206-B 207-B 208-C 209-C 210-D 211-B 212-A 213-D 214-A 215-B 216-B 217-A 218-C 219-A 220-C
 221-C 222-C 223-B 224-B 225-B 226-B 227-B 228-C 229-B 230-B 231-B 232-B 233-B 234-B 235-B 236-B 237-C 238-A 239-B 240-B
 241-A 242-A 243-B 244-C 245-C 246-B 247-A 248-C 249-B 250-C 251-B 252-A 253-B 254-A 255-C 256-C 257-A 258-B 259-B 260-A
 261-B 262-C 263-A 264-B 265-B 266-B 267-B 268-B 269-B 270-A 271-A 272-B 273-B 274-A 275-B 276-A 277-A 278-C 279-A 280-B
 281-B 282-B 283-C 284-D 285-B 286-A 287-B 288-A 289-B 290-B 291-A 292-B 293-C 294-B 295-A 296-A 297-A 298-A 299-B 300-A
 301-A 302-A 303-A 304-A 305-A 306-A 307-A 308-A 309-A 310-A 311-A 312-A 313-C 314-A 315-A 316-A 317-A 318-B 319-A 320-A
 321-A 322-A 323-A 324-B 325-A 326-B 327-C 328-A 329-A 330-A 331-A 332-A 333-B 334-A 335-C 336-B 337-A 338-A 339-A 340-B
 341-C 342-A 343-B 344-A 345-D 346-A 347-C 348-B 349-B 350-B 351-A 352-B 353-B 354-B 355-A 356-C 357-B 358-B 359-B 360-B
 361-B 362-A 363-B 364-B 365-B 366-B

Source & Provenance Notes

Primary course materials

- All previously uploaded CSE-306/MMA lecture slides and laboratory sheets used in the original 226-question master note.
- New source added in this edition: "MMA quiz.pdf" (35-page scan), especially trainer-kit workflow, KIT/PC mode, machine-code entry, LCD code explanation, busy-flag subroutine, blinking code, and dot-matrix program.

Web MCQ-pattern research (paraphrased, syllabus-filtered)

- GeeksforGeeks - 8086 Instruction Set: <https://www.geeksforgeeks.org/electronics-engineering/8086-instruction-set/>
- TutorialsPoint - 8086 Addressing Modes: https://www.tutorialspoint.com/microprocessor/microprocessor_8086_addressing_modes.htm
- Sanfoundry - 8086 Addressing Modes MCQs: <https://www.sanfoundry.com/microprocessors-mcqs-addressing-modes-8086/>
- Sanfoundry - 8086/8088 Instruction Set MCQs: <https://www.sanfoundry.com/microprocessors-mcqs-instruction-set-8086-8088-1/>
- Sanfoundry - Stack MCQs: <https://www.sanfoundry.com/microprocessors-mcqs-stack/>

Important scope rule

The web was used only to identify standard question patterns for concepts already taught in the uploaded materials. Out-of-syllabus topics were excluded. Web questions were not copied verbatim.